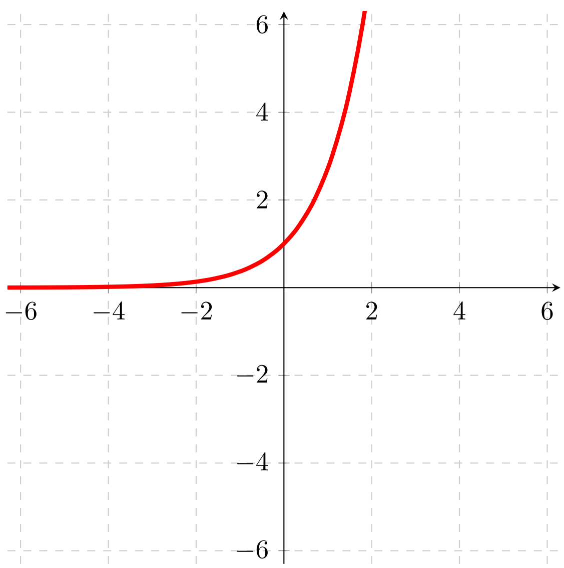
Problem 1

Problem 1

Problem 1 *The graph of $g(x) = e^x$ is given below.*

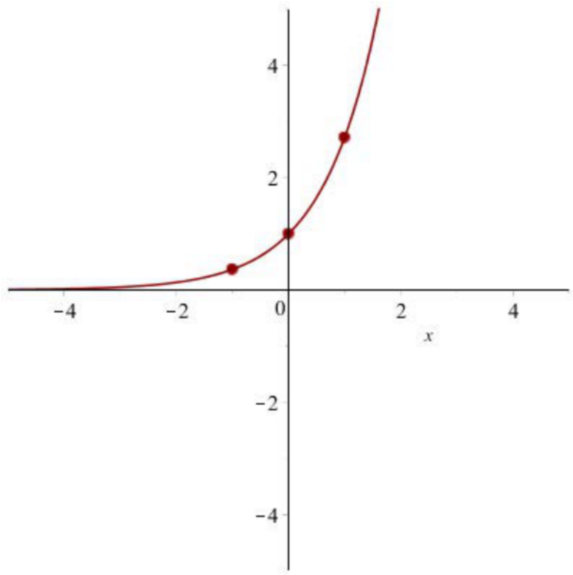


(a) *Find the domain and range of g.*

Explanation. Domain: $(-\infty, \infty)$, Range: $(0, \infty)$

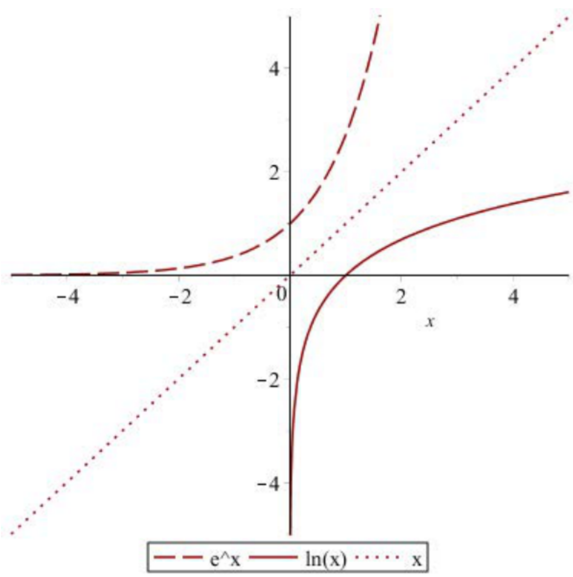
(b) *Find the values of $g(1), g(0), g(-1)$ and plot the points $(1, g(1)), (0, g(0))$, and $(-1, g(-1))$ on the graph of $g(x) = e^x$.*

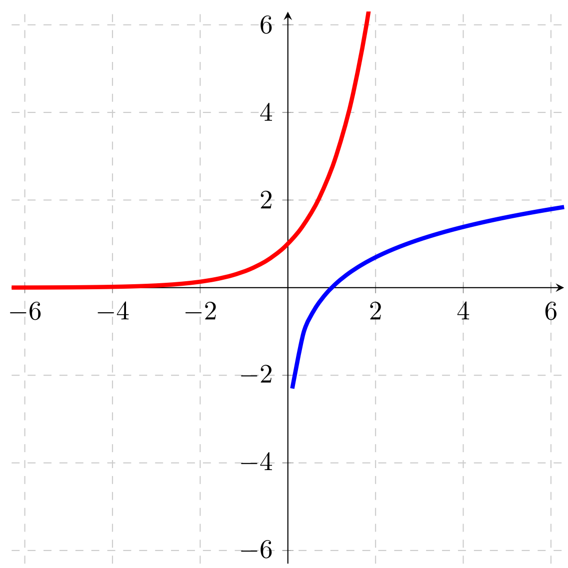
Explanation. $g(1) = e^1 = e$, $g(0) = e^0 = 1$, and $g(-1) = e^{-1} = \dfrac{1}{e}$. These are standard values of the famous function $g(x) = e^x$ that you should know.



(c) *Graph $h(x) = \ln(x)$ on the same axis as $g(x) = e^x$.*

Explanation. Recall: $\ln(x)$ is the inverse of e^x. To find the graph of $\ln(x)$ we reflect the graph of e^x over the line $y = x$. Since the points $\left(-1, \dfrac{1}{e}\right)$, $(0,1)$, $(1,e)$ are on $y = e^x = g(x)$, the points $\left(\dfrac{1}{e}, -1\right)$, $(1,0)$, $(e,1)$ are on the graph of $y = \ln(x) = g^{-1}(x) = h(x)$





(d) *Find the domain and range of* h*.*

Explanation. The domain of $h(x) = \ln(x)$ is $(0, \infty)$. The range is $(-\infty, \infty)$.

(e) *Find the values of* $h(1), h(0), h(-1), h(e), h\left(\frac{1}{e}\right)$*, or say* x *not in the domain.*

Explanation. $h(1) = \ln(1) = 0$, $h(0)$ is not defined because 0 is not in the domain of $\ln(x)$. $h(-1)$ is not defined because -1 is not in the domain of $\ln(x)$. $h(e) = \ln(e) = 1$, and $h\left(\frac{1}{e}\right) = \ln\left(\frac{1}{e}\right) = \ln\left(e^{-1}\right) = -1\ln(e) = -1$. These are standard values of the famous function $h(x) = \ln(x)$ that you should know.